

## DUSTRUNNER · PROBLEM STATEMENT

# DustRunner problem statement

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## DOCUMENT CONTROL

|             |                                         |         |                |
|-------------|-----------------------------------------|---------|----------------|
| Document ID | DRN-PRB-001                             | Version | 0.5            |
| Status      | <b>Draft</b>                            | Date    | 2026-10-02     |
| Author      | Amish Chadha                            | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept           |         |                |
| Repository  | github.com/BoujeeEnjinia1701/dustrunner |         |                |
| Source      | docs/01-problem.md @ uncommitted        |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                                |
|---------|------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 0.5     | 2026-10-02 | Amish Chadha | First candidate pilot site and co-design partner, as decided by Amish on 2026-10-02                                                   |
| 0.4     | 2026-09-26 | Amish Chadha | Stronger sources                                                                                                                      |
| 0.3     | 2026-09-25 | Amish Chadha | Apply DRN-DDR-001 (problem line reworded to a monthly loss in the dusty season; first table format decided; pilot partner still open) |
| 0.2     | 2026-09-25 | Amish Chadha | Populate to TRL 2 (problem, users, context, constraints, prior work)                                                                  |
| 0.1     | 2026-09-24 | Amish Chadha | Initial scaffold                                                                                                                      |

# DustRunner problem statement

Soiling cuts output of PV arrays in dusty regions by up to 15 to 25% a month in the dusty season, and water for washing is scarce. Utility-scale plants solve this with water-free cleaning robots, but small and mid-size ground-mounted arrays of tens of kilowatts, which are common on farms, mini-grids and commercial sites in the same regions, are still washed by hand with water or left dirty.

## ■ The problem

Dust settles on PV glass every day. In desert climates the loss grows quickly between cleanings. Qatar's QEERI test facility measured a soiling rate of about 0.49 % per day on fixed-tilt modules, about 13 % a month without rain or cleaning, and losses of up to 60 % over long uncleaned periods ([PV Tech, "Challenges of PV soiling in desert climates"](#)). A 2026 study for Arar, Saudi Arabia, assumed monthly losses of about 4 % in winter rising to 15 % in June and July ([Alharbi, Energies 19\(18\) 4373](#)). The "up to 15 to 25 % a month" in the summary above is therefore the loss of an array left uncleaned for a month or more in the dusty season, not a year-round average.

Worldwide, soiling cost at least 3 to 4 % of solar production in 2018, about 3 to 5 billion euros of revenue, with 4 to 7 % projected for 2023 ([Ilse et al., Joule 3, 2303 to 2321, 2019; open access PDF](#)).

Washing with water works but uses water where it is scarcest. Field trials found about 3.5 L to more than 10 L of water used to clean and rinse a single module ([Polywater, "Water consumption in PV panel cleaning"](#)). Manual cleaning also needs labor, so small owners stretch the interval: the cost-optimal manual interval came out at about 60 days, and longer at low tariffs ([Alharbi, 2026](#)), and the array runs dirty for most of that time.

The underlying gap is that each manual cleaning is expensive, so it happens rarely. A small robot that cleans dry every day changes that trade-off, provided it is cheap enough for one row and does not damage the glass.

## ■ Users and context

| User                                  | Need                                                                                                 | Context                                                                                                 |
|---------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Owner of a small ground-mounted array | More energy from the same array without buying water or hiring cleaners                              | Farms (solar irrigation pumping), mini-grids, schools and clinics, small commercial sites; 10 to 200 kW |
| O&M technician or site caretaker      | A machine that runs itself, is easy to install on an existing row and is repairable with local parts | Visits the site weekly or less                                                                          |
| Mini-grid or community solar operator | Predictable yield and lower operating cost in the dry season                                         | Arrays of several rows, remote sites                                                                    |
| Open hardware community               | A documented, buildable reference design to adapt to other module and table sizes                    | Makerspaces, university labs, technical colleges                                                        |

Operating context assumed for the concept:

- **Climate:** hot, dry and dusty (for example the Middle East and North Africa, the Sahel, northwest India and the US Southwest). Ambient 0 to 50 °C, glass up to about 75 °C, nightly dew possible in coastal and desert areas.

- **Arrays:** fixed-tilt ground-mounted tables, one module high in portrait (1P), tilt 10 to 35°, rows 20 to 100 m long. Modules of about 2,278 x 1,134 mm with 30 to 35 mm aluminum frames are now common ([DAS Solar 590 W example](#)).
- **Dust:** mostly dry, loose mineral dust between rain events. Cemented dust after dew, bird droppings and lichen are outside what dry brushing can remove.

## ■ Prior work

- **Ecoppia** (Israel, India, United States): water-free robots that use microfiber elements and airflow, charge from their own onboard solar module and clean utility-scale rows nightly and autonomously ([Ecoppia technology](#)). Sold to large plants as a service.
- **NOMADD** (Saudi Arabia): the "NO-water Mechanical Automated Dusting Device", designed to clean every panel as often as daily in desert plants, at about 10 US cents per watt of added cost ([RenewEconomy](#)).
- **SolarCleano F1** (Luxembourg): an 82 kg operator-moved robot with 1.2 m helical brushes, dry or wet, up to 1,200 m<sup>2</sup>/h, able to cross gaps up to 70 cm ([SolarCleano](#)). Suited to contractors, not to a single small row.
- **Academic prototypes:** for example a tracked robot with two helical brushes and ultrasonic sensing for desert PV ([Antonelli et al., Mechatronics, 2020](#)).
- **Robotic cleaning cost** on utility-scale plants is about 2.4 to 8.2 € per m<sup>2</sup> of module area ([Ilse et al., 2019](#)).

The gap DustRunner targets is an open, garage-buildable robot for one small row that installs without rails and costs less than one or two years of the energy it recovers.

## ■ Constraints

- Garage-buildable prototype, \$500 USD ( `budget_usd` ) for one robot, its dock and its anemometer, using off-the-shelf motors, aluminum sections and hobby electronics.
- Must not void or endanger module warranties. Manufacturers forbid abrasive cleaning and some require prior approval of motorized cleaning tools; First Solar, for example, reviews automated tools and gives no warranty for modules damaged by cleaning ([First Solar module cleaning guidelines](#)). Abrasion of anti-reflective coatings by robots is still being studied in the field ([PV Tech; Ilse et al., 2019](#)).
- No drilling of modules or of the mounting structure; only clamp-on parts.
- Works on an energized array. PV strings can carry DC voltages of several hundred volts, so the robot must never touch cables, connectors or junction boxes.
- No water and no consumables other than replaceable microfiber.

**Safety:** DustRunner is unattended moving machinery with a lithium iron phosphate battery, working on an energized PV array on hot glass up to about 1.6 m above the ground. The hazards and the measures against them are in DRN-PRC-001, Safety.

## ■ Out of scope

- Utility-scale plants, single-axis trackers and rooftop arrays.
- Automatic transfer from row to row (one robot per row, or carried by hand).
- Wet cleaning, bird droppings and cemented soiling.

- A commercial cleaning service or fleet software.

## ■ Open questions

- Which pilot site and co-design partner to work with first (a solar pumping farm, a mini-grid operator or a university test array)? Decided by Amish on 2026-10-02: a university or national lab outdoor PV test site in a dusty climate, with Arizona State University's Photovoltaic Reliability Laboratory in Mesa as the first candidate to approach (DRN-DEC-001).
- First module and table format: decided by Amish on 2026-09-25 as 1P portrait tables of 2,278 mm modules (DRN-DDR-001, D10). The visible frame flange width of target modules still needs a survey (DRN-CAL-001, C12).
- Is daily dry cleaning effective on the local dust, or does dew cementation limit it? This needs site dust data and later testing.